Principal component analysis

How samples relate in a low-dimensional view of the chosen matrix, and what dominates the variation when they do not.

AT A GLANCE

Where	Quality control → PCA
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, at least two samples, and enough features observed in *every* sample.
Data source	The only quality-control page with a choice — Data, Filtered Data or Imputed Data — because a projection cannot use a feature that is missing anywhere, so which matrix goes in is a scientific decision.
Methods	PCA, UMAP and t-SNE. Only PCA offers three dimensions and only PCA labels its axes with explained variance.
Two renderers	**Type** Normal draws a static figure; Interactive opens a zoomable, rotatable plot in a new tab.
Table	None. The feature count the projection rests on is reported with the controls instead.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

PCA summarises the differences between samples using a few new axes. Each axis combines the retained features: PC1 captures the largest remaining variance, PC2 the next largest in a perpendicular direction, and so on. Nearby samples have similar scores on the displayed components. Samples far apart differ in the patterns represented by those components, but may still be similar in other respects.

Use the plot to ask whether variation follows condition, batch or an individual sample. Separation is not a significance test and does not identify its cause. A treatment effect need not dominate PC1, and overlapping conditions can still contain differentially abundant proteins.

The axes are quantitative component scores. Their interpretation depends on feature scaling; their signs are arbitrary, so a mirrored PCA can represent the same result. The variance percentages refer to the matrix actually analysed, not every row of the uploaded file.

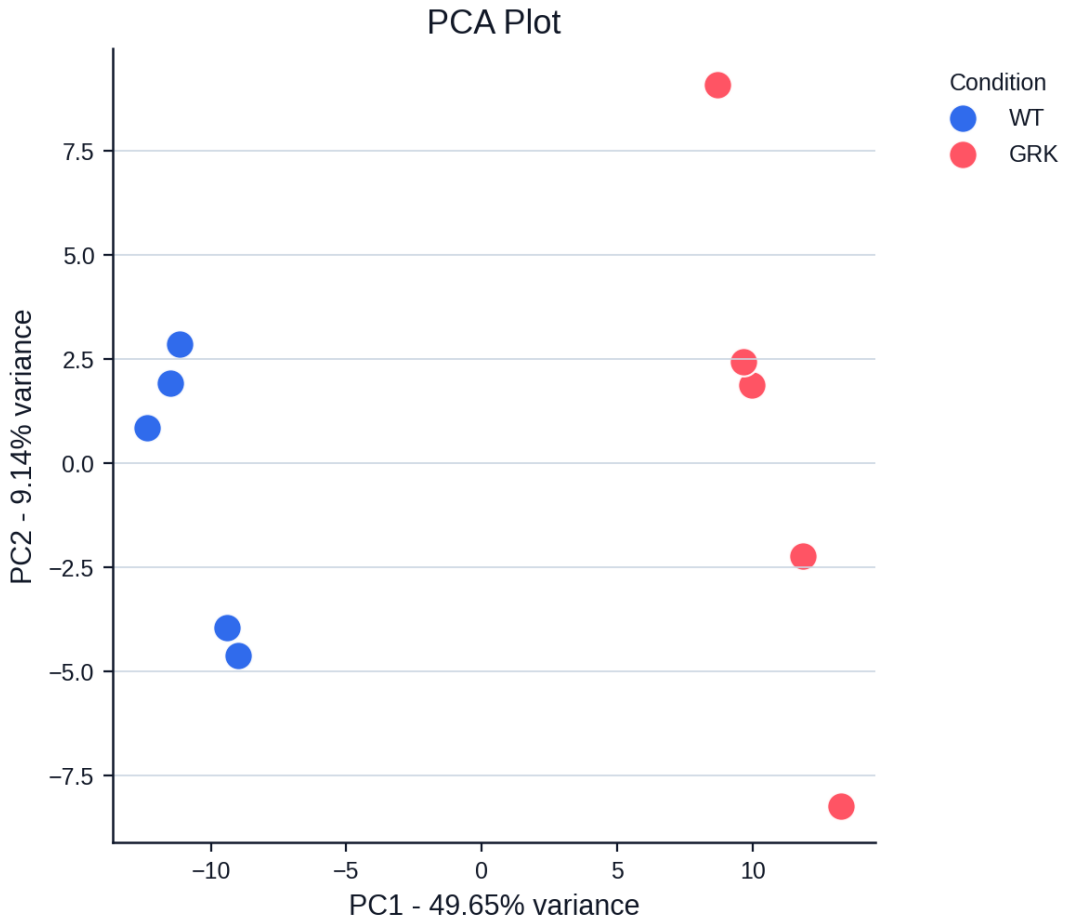


Figure 1.1. PCA of the protein-level matrix of the example dataset, in two dimensions and without ellipses: ten points, five WT on the left and five GRK on the right, separated along PC1 and mixed along PC2. PC1 carries 49.65 % of the variance in the reduced matrix and PC2 9.14 %. These values describe this saved example, not a target that another experiment should reproduce.

Reading it

- One point, one sample — never a feature. The features are what define the axes; the samples are what is plotted on them.
- The axis labels. For PCA they read PC1 - 49.65% variance: the share of the total variance this component accounts for, so the two percentages together say how much of the dataset the picture represents. UMAP and t-SNE axes carry no percentage, because neither preserves variance.
- The ellipse, with **Add Ellipses (2D)** on: the *smallest ellipse containing every point of that group*. It summarises where the group lies; it is not a confidence region, and it can only grow as replicates are added.
- The colour: the condition, from the annotation and shared with every other plot in the session, or the cluster when **Color By** says so.

> **EASY TO GET WRONG**
>
> This implementation uses complete, non-constant features. Features with missing values or no variation are excluded, and the page reports the usable count. This can favour consistently detected features. A small retained set is a reason to inspect missingness, not an automatic reason to impute. If an imputed matrix is used, compare it with the observed-data result: the positions may partly reflect the imputation assumptions.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to project: protein, phosphosite or phosphoprotein.	Protein	page
Data Source	Which matrix to project: Data as uploaded, Filtered Data or Imputed Data. Only the ones that exist are offered.	the most processed matrix available	page
Method	PCA, UMAP or t-SNE. Changing it resets **Feature scaling** to that method's own default, and returns **Dimensions** to 2D and **Clustering** to None where those do not apply.	PCA	page

Controls on this page (continued)

Control	What it does	Default	Where
Feature scaling	Centred subtracts each feature's mean across the samples and stops there, so a feature contributes in proportion to its variance. Unit variance also divides by the standard deviation, giving every feature equal weight.	Centred for PCA, Unit variance for UMAP and t-SNE	page
Type	Normal draws a static figure that can be downloaded. Interactive builds a Plotly figure with zoom, rotation and per-sample hover, opened in a new tab.	Normal	page
Dimensions	2D plots PC1 against PC2, 3D adds PC3. **Method** PCA only.	2D	page
Add Ellipses (2D)	Draws the minimum enclosing ellipse of each group. Offered in two dimensions only.	off	page
Clustering	None, KMeans, Agglomerative or DBSCAN. It is run on the two embedding coordinates rather than on the original features, so it clusters what the picture shows rather than what the data holds. Offered for UMAP and t-SNE only.	None	page
Color By	Condition, or Cluster once a clustering method has been chosen.	Condition	page
Cluster Count	How many clusters to form. KMeans and Agglomerative only; held between 2 and 20.	3	page
DBSCAN Eps, DBSCAN Min Samples	The neighbourhood radius in embedding coordinates, and the number of points that makes one. DBSCAN only.	0.5, 2	page
UMAP Neighbors, UMAP Min Dist	How much neighbourhood structure UMAP preserves, and how tightly it may pack points. Held between 2 and 200, and between 0 and 1.	15, 0.1	page
t-SNE Perplexity, t-SNE Learning Rate	The effective neighbourhood size, capped at one less than the number of samples, and the step size of the optimisation.	30, 200	page
Random State	Seeds UMAP and t-SNE so that the same figure is reproduced exactly. Neither is deterministic without it.	42	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure, shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF. Normal mode only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Points	Diameter in points, shape, opacity and outline — for all points at once, or for one condition at a time.	6 static, 9.6 interactive	appearance
Condition names, Condition colors	One display name and one colour per group. With **Color By** set to Cluster the rows address the clusters instead.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written from the method: PCA Plot, UMAP Plot, t-SNE Plot, prefixed 3D in three dimensions.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its heading follows **Color By**.	shown	appearance
X axis, Y axis	Overwriting an axis label removes the explained-variance percentage with it.	plot's own	appearance
Frame and grid	The two renderers start from different conventions: spines on and no grid for the static figure, grid on and no spines for the interactive one.	by renderer	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

1. The matrix is the one **Data Source** names. Non-numeric and non-finite entries are read as NA.
2. Every feature with an NA in any sample is dropped. If fewer complete features remain than the projection has dimensions, no figure is drawn and the notice says so.
3. The matrix is transposed — the samples become the rows and the features the

columns — and any feature that takes the same value in every sample is dropped, having no variation to contribute.

4. Each feature is centred on its mean across the samples. With **Feature scaling** set to Unit variance it is then divided by its standard deviation across the samples, taken with $n-1$ in the denominator as R's prcomp(scale. = TRUE) does; a feature whose standard deviation is zero is divided by one instead of being lost.

5. For PCA: the singular value decomposition of that matrix, $X = U\Sigma V^{\top}$. The sample coordinates are the first components of $U\Sigma$, and the label on component k is $\sigma_k^2 / \sum_l \sigma_l^2 \times 100\,\%$ — a share of the variance *in the reduced matrix*, after incomplete and constant features were dropped, not of the uploaded dataset.

6. For UMAP and t-SNE: the same prepared matrix goes to the respective algorithm with the parameters set on the page. Neither produces an ordered decomposition, so the axes are numbered and carry no percentage.

THE ELLIPSE

For each group, the smallest ellipse containing all of its points is fitted by Khachiyan's algorithm and drawn at 15 % opacity in the group's own colour. The fit needs at least three points that span the plane. Three points on a line, or two at the same position and a third, leave it undefined, and an ellipse whose shorter axis collapses would draw as a line — which reads as a region of zero area rather than as the absence of one. In either case that group gets no ellipse, its points are drawn as usual, the other groups are unaffected, and a note with the controls names the groups skipped. The figure itself never says so: a plot that leaves the tool should carry no caption about how the tool drew it.

For example, PC1 at 40 % and PC2 at 10 % display half the variance in the analysed matrix. They leave the other half out of view. Separation on a 4 % component can still be biologically relevant; the percentage measures variance, not importance or confidence. Compare sample positions with the experimental design, coverage and intensity distributions before excluding a sample. Biological replicates need not be as similar as technical repeats.

UMAP and t-SNE answer a related but different question about neighbourhood structure. Their axes are not fold changes or percentages of variance. Cluster area and the distance between separate islands should not be read as quantitative measures of biological difference. Check whether an apparent pattern persists across reasonable settings; clustering an embedding is exploratory, not an independent validation of the groups.

Edge cases

- Two samples are needed for a two-dimensional projection and three for a three-dimensional one; below that no figure is drawn.
- Selecting Imputed Data can increase the complete-feature count, but remaining non-finite

or constant features are still excluded. Imputation is not a guarantee that every row is usable.

- If every complete feature is constant, there is nothing to decompose and the page reports that instead.
- Ellipses exist in two dimensions only; the switch is hidden in 3D. A condition with one or two samples cannot bound one and is named among the skipped groups.
- UMAP requires the umap-learn package and t-SNE requires scikit-learn. If one is absent the page says which, and how to install it.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = pca_plot(data, meta)
```

R

```
fig <- pca_plot(data, meta)
```

add_ellipses=True and plot_3d=True are the page's **Add Ellipses (2D)** and **Dimensions** switches, and scaling takes the same "auto", "centred" and "unit_variance" the **Feature scaling** selector offers. There is no data-source argument: which matrix is projected is decided by which frame is handed in, so an imputed projection means passing the imputed frame.

5 Exporting

In Normal mode the download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. In Interactive mode there is no image to download: the plot opens in its own tab, where Plotly's camera button saves what is on screen at screen resolution. For a figure that will be scaled or edited, use Normal and PDF.

6 Further reading

For an accessible introduction, see Wikipedia: principal component analysis. For nonlinear projections, Wattenberg, Viégas and Johnson, How to Use t-SNE Effectively illustrates why island size and spacing need care. These explain the methods; the controls and row exclusions above describe CoPYlot.

Behind this page

Function	What it produces
pca_plot	The static projection, in two or three dimensions, with the optional ellipses
pca_plot_interactive	The interactive one. The package function takes no scaling argument and always projects the centred matrix; the page applies **Feature scaling** in both modes
resolve_feature_scaling	Which scaling a method runs on
fitted_group_ellipse	One group's ellipse, or nothing when its points do not span the plane